

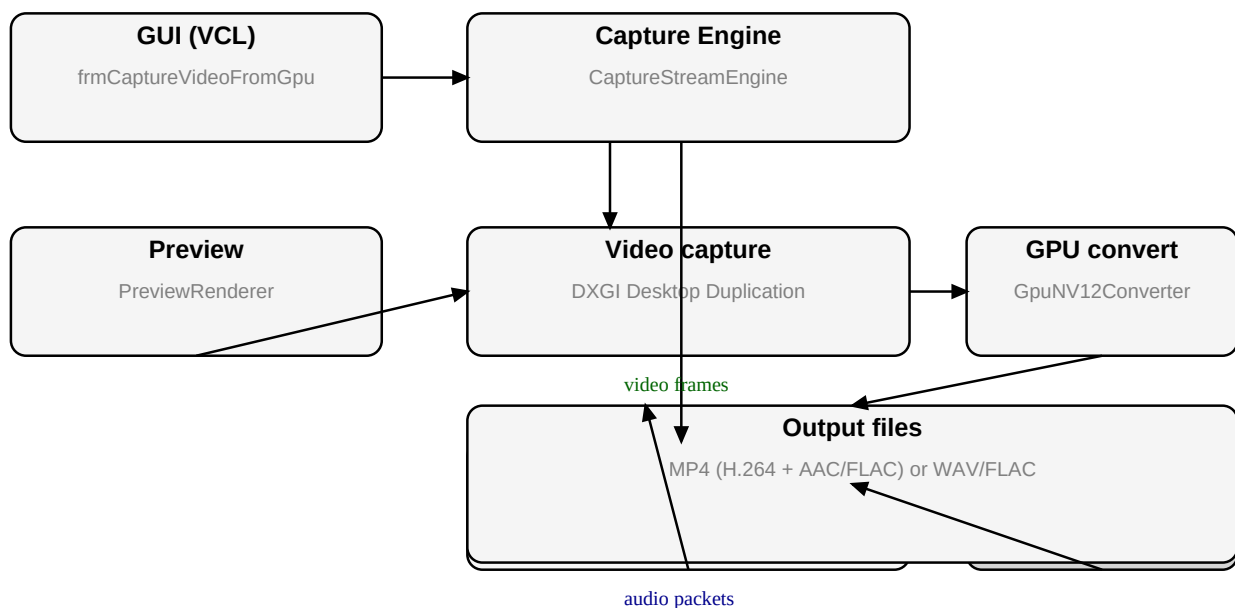
CaptureVideoFromGPU

Application structure, key methods, and data flow (beginner-friendly)

What it does

Captures a selected monitor using DXGI Desktop Duplication, optionally captures system audio via WASAPI loopback, shows a live preview, and writes the result to a file (typically MP4 with H.264 video and AAC or FLAC audio). Audio-only recording can be done separately to WAV or FLAC using the dedicated recorder unit.

Big picture flow



Main units and responsibilities

Unit	Role
frmCaptureVideoFromGpu.pas / .dfm	VCL GUI: device selection, preview panel, start/stop, logging.
CaptureStreamEngine.pas	Core engine: D3D11 + Desktop Duplication, timing, sample creation, sink writer, threads.
PreviewRenderer.pas	On-screen preview (swapchain rendering into a panel HWND).
GpuNV12Converter.pas	GPU conversion (desktop BGRA/RGBA -> NV12 for H.264 input).
WasapiLoopbackCapture.pas	WASAPI loopback capture thread; raises OnData(Buffer, NumFrames, WaveFormat).
Helpers.pas	Shared utilities (HRESULT checking, small helpers).
LoopbackAudioOnlyRecorder.pas (optional)	Standalone audio-only recording to WAV/FLAC (simple for teaching).

Key methods (what to look at first)

Area	Method	Purpose
GUI	TfrmCapture.btnStartClick	Read UI settings, configure audio options, call engine StartCapture.
GUI	TfrmCapture.btnStopClick	Call engine StopCapture and finalize files.
Engine	TCaptureStreamEngine.StartCapture	Initialize devices, create sink writer streams, start threads.
Engine	TCaptureStreamEngine.StopCapture	Stop threads, stop loopback, finalize sink writer.
Video	VideoLoop	Acquire frames via Desktop Duplication, preview, convert to NV12, queue samples.
Audio	OnWasapiData	Receive loopback packets, convert/pack PCM, queue audio samples.
Writer	SinkWriterLoop	Single place that calls IMFSinkWriter.WriteSample for audio/video.
WASAPI	TWasapiLoopbackCapture.InitClient	Select endpoint (default or by ID) and init IAudioClient for loopback.
WASAPI	CaptureThreadProc	Read packets and call OnData callback.

Lifecycle notes (beginner rules of thumb)

- Configure audio BEFORE StartCapture (streams are created when the sink writer is created).
- StopCapture must run to finalize the file (sink writer finalize and proper headers).
- Keep all IMFSinkWriter.WriteSample calls in one thread (writer loop) for stability.
- Loopback audio usually arrives as WAVE_FORMAT_EXTENSIBLE float32; convert to PCM16 for simple, compatible output.
- If you rely on the default render endpoint, eMultimedia often matches what you actually hear better than eConsole.

Troubleshooting checklist (fast)

- No audio: confirm WasapiLoopbackCapture.OnData fires and that audio samples are enqueued and written.
- 0-byte WAV: Stop was not called (WAV header sizes not patched) or wrong output path.
- Hollow sound: capture the correct default endpoint/role (try eMultimedia) or pass the explicit device ID.